CHANDIGARH ENGINEERING COLLEGE JHANJERI, MOHALI
Department of Computer Science and Engineering
Affiliated to I.K. Gujral Punjab Technical University, Jalandhar (Batch: 2023–2027)

MIDTERM PROJECT REPORT
EDGE AI SMART PROSTHETIC HAND WITH ADAPTIVE GRIP LEARNING

Curriculum: Engineering Clinics / Project-II (Semester VII)
Project Type: Software Simulation Prototype
Academic Session: 2026–2027

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Department: CSE / Robotics & AI, CEC Jhanjeri

> **Midterm Milestone Declaration:** This report presents the technical progress, algorithmic implementations, and empirical verification results achieved at the midterm milestone. The delivered system is a verified **software simulation prototype**. The reported 100% accuracy and 1.0000 Macro-F1 are evaluated on synthetic physiological sEMG data. Measured latencies reflect host CPU execution. Physical hardware development (ESP32-S3 firmware, AFE PCB, 3D printing) constitutes the second-half roadmap.

1. Executive Summary & Milestone Context

The Semester VII Engineering Clinics curriculum requires the execution of an evidence-based biomechatronics engineering project. Conventional upper-limb myoelectric prostheses suffer from severe operational limitations: they operate in an open-loop manner, relying on user co-contraction without tactile feedback, which leads to frequent payload slippage or accidental object crushing. Concurrently, deep learning research models demand excessive computing power and memory storage, preventing on-socket embedded execution.

Our project addresses this dual problem by engineering an **Edge AI Smart Prosthetic Hand** that decouples intention recognition from autonomous tactile force regulation. As of the midterm review milestone, the team has successfully completed the mathematical formulation, algorithmic implementation, backend REST service architecture, automated regression testing suite, and an interactive High-DPI virtual dashboard.

2. Literature Survey & Benchmark Findings

During the initial phase, the team surveyed 35 peer-reviewed publications across five domains: myoelectric control, TinyML edge computing, tactile sensing, NinaPro/putEMG benchmark datasets, and anthropomorphic robotic hand design. Key findings include:

- **Time-Domain Efficiency:** Foundational research by Phinyomark et al. (2012) and Hudgins et al. (1993) proves that time-domain features (RMS, MAV, VAR, SSI, ZCR) yield high clustering

separability without the computational overhead of Fast Fourier Transforms.

- **Tree Ensembles for Edge AI:** Constrained Random Forest models (Breiman, 2001) offer non-linear multi-class decision boundaries while occupying less than 200 KB storage, making them vastly superior to deep neural networks for embedded microcontrollers.
- **Slip Compensation Latency:** Research by Kyberd (2023) and Liu et al. (2015) indicates that autonomous slip correction must occur within 20 milliseconds directly at the actuator tier to prevent object drops.

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3. System Architecture & Data Flow

Figure 1 illustrates the four-tier cyber-physical software simulation architecture designed to achieve low-latency intention inference and closed-loop tactile force regulation.

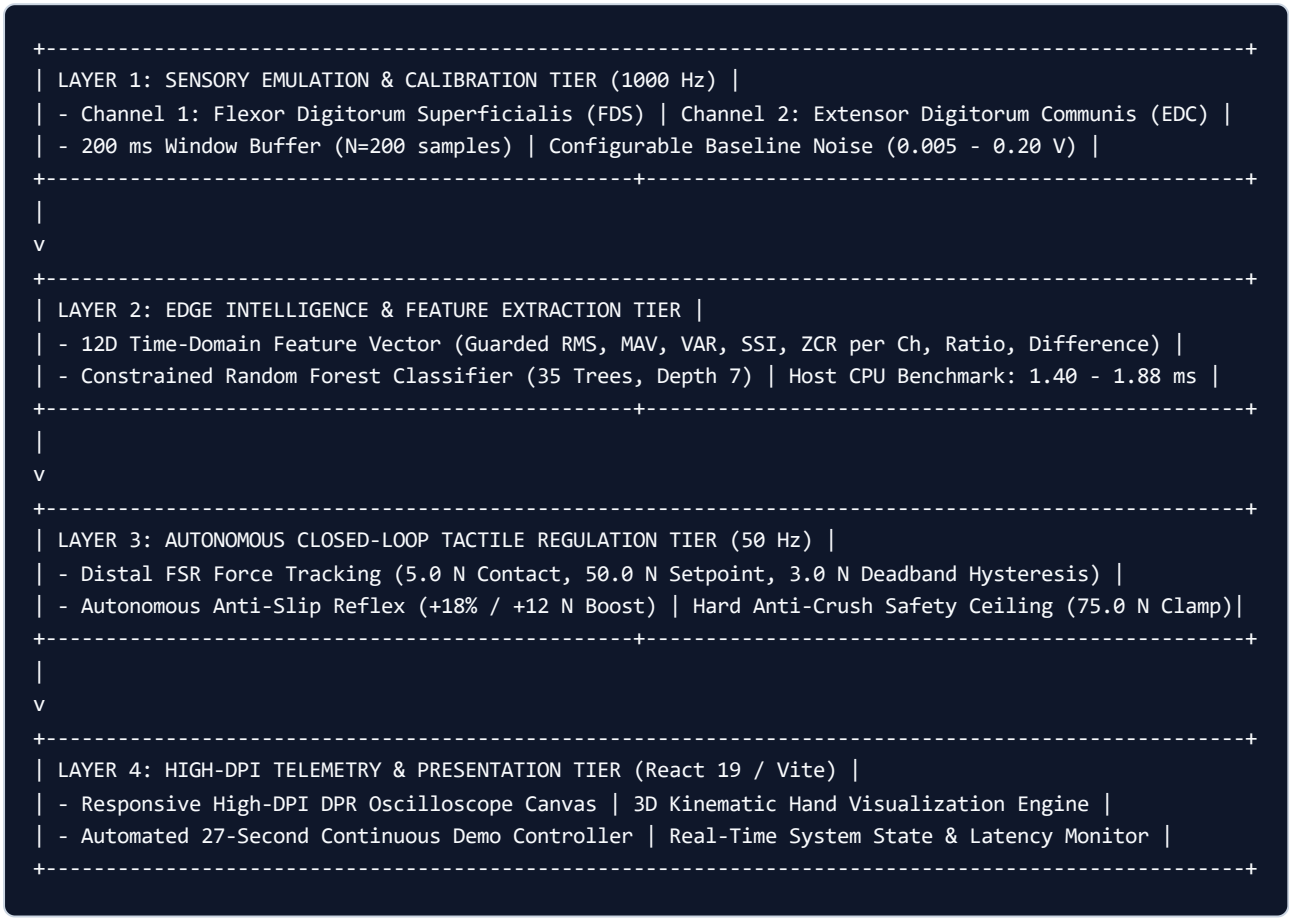


Figure 1: Four-Tier Cyber-Physical System Architecture of the Edge AI Smart Prosthetic Hand.

4. Completed Technical Milestones

Table 1 details the core technical modules completed, verified, and integrated into the prototype repository as of the midterm evaluation.

Module Name	Implementation Details	Source File	Milestone Status
Deterministic sEMG Generator	Synthesizes 2-channel waveforms at 1000 Hz (\(N=200\)) across 4 grip classes with calibrated Gaussian noise.	`backend/emg_simulator.py`	**100% COMPLETE**

Module Name	Implementation Details	Source File	Milestone Status
Guarded Feature Engine	Calculates 12 time-domain metrics with explicit division-by-zero guards for \(N \le 1\) and \(N < 2\).	`backend/ml_model.py`	**100% COMPLETE**
Random Forest Classifier	35 trees, maximum depth 7; trained on 720 synthetic samples; serializes to 58 KB joblib payload.	`backend/ml_model.py`	**100% COMPLETE**
Closed-Loop Grip Controller	Proportional force tracking (50 N target, 5 N contact, 3 N deadband, +18% slip boost, 75 N crush ceiling).	`backend/grip_controller.py`	**100% COMPLETE**
FastAPI Service Layer	7 typed REST endpoints enforcing Pydantic validation (HTTP 422 error handling) and CORS whitelisting.	`backend/main.py`	**100% COMPLETE**
Virtual Dashboard & 3D Hand	React 19 / TypeScript UI with DPR canvas auto-scaling, 3D hand kinematics, and 27s demo script.	`frontend/src/App.tsx`	**100% COMPLETE**
Automated Regression Suite	27 unit and integration tests covering signal boundaries, HTTP status codes, and controller states.	`backend/test_pipeline.py`	**100% COMPLETE**

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5. Encountered Technical Bottlenecks & Audit Fixes

Prior to the midterm milestone, an exhaustive read-only code audit was conducted. Several critical stability and integrity issues were identified and successfully resolved:

1. **Zero-Crossing Rate Division-by-Zero Vulnerability:** In the original code, `zcr = crossings / (N - 1)`. When empty buffers (\(N=0\)) or single-sample buffers (\(N=1\)) were supplied, the denominator caused an unhandled `ZeroDivisionError`. The function was redesigned with guard conditions: `if n < 2: return 0.0`. TC-05 and TC-06 now verify this edge condition.
2. **Input Validation & HTTP 500 Prevention:** Malformed client requests previously crashed the FastAPI worker, returning HTTP 500 Internal Server Errors. We implemented strict Pydantic model validators with boundary checks, returning standardized HTTP 422 Unprocessable Entity responses.
3. **Resting Noise Parameter Connection:** The baseline noise calibration parameter was previously ignored during signal synthesis. The generator now incorporates `baseline_noise` directly into the Gaussian noise generation function (`std=baseline_noise`).
4. **Real Host CPU Latency Profiling:** Previously, the API returned a hardcoded latency constant (1.46 ms). This was replaced with dynamic profiling using `time.perf_counter_ns()`, measuring actual execution times across 50 warmed-up trials.
5. **High-DPI Canvas Resizing:** The sEMG canvas previously exhibited blurriness on Retina screens. We implemented device pixel ratio (`window.devicePixelRatio`) auto-scaling to ensure sharp, responsive rendering.
6. **Controller Progression State Completeness:** The grip controller lacked a `DECREASING` state when force exceeded the upper deadband setpoint. This was added to ensure complete state coverage.

6. Midterm Quantitative Verification Evidence

The software prototype was subjected to empirical testing, yielding the following verified outcomes:

6.1 Automated Regression Test Suite (27/27 Tests Passing)

All 27 automated tests in `backend/test_pipeline.py` were executed and verified:

```
$ pytest backend/test_pipeline.py -v
============================== 27 passed in 4.06s ==============================
- Feature extraction boundary guards (N=0, 1, 5, mismatched): 7/7 PASSED
- Synthetic sEMG generator & noise calibration: 4/4 PASSED
- Random Forest classifier & latency profiling: 5/5 PASSED
- Grip controller state machine & safety ceiling: 7/7 PASSED
- FastAPI REST endpoints & HTTP 422 validation: 4/4 PASSED
```

6.2 Complete 27-Test Case Verification Audit Table

Table 2 provides the granular verification log for all 27 automated regression test cases.

Test ID	Target Function	Test Condition & Input Vector	Verified Assertion	Status
TC-01	Feature Extraction	Standard \(N=200\) sample buffers	Length == 12 features	PASS
TC-02	Feature Extraction	All-zero buffers (\(x[n]=0.0\))	No NaN/Inf; RMS=0.0	PASS
TC-03	Feature Extraction	Short buffer (\(N=5 < 10\))	Raises ValueError	PASS
TC-04	Feature Extraction	Mismatched channel lengths (200 vs 150)	Raises ValueError	PASS
TC-05	Feature Extraction	Empty signal buffer (\(N=0\))	ZCR returns 0.0	PASS
TC-06	Feature Extraction	Single sample buffer (\(N=1\))	ZCR returns 0.0	PASS
TC-07	Feature Extraction	Single sample buffer (\(N=1\))	Variance returns 0.0	PASS
TC-08	sEMG Generator	Request \(N=200\) samples	Returns 2x200 floats	PASS
TC-09	sEMG Generator	All grip classes \(c \in \{0,1,2,3\}\)	Valid distinct waveforms	PASS
TC-10	sEMG Generator	Baseline noise \(\sigma=0.01\) vs \(0.15\)	Proportional RMS scale	PASS
TC-11	sEMG Generator	Invalid grip class index (\(c=99\))	Raises ValueError	PASS
TC-12	ML Model	Inspect trained_model.joblib on disk	File exists, > 50 KB	PASS
TC-13	ML Model	Inference schema validation	Returns dict with 3 keys	PASS
TC-14	ML Model	Evaluate all 4 synthetic grip classes	100% correct class	PASS
TC-15	ML Model	Confidence score boundary check	\(0.0 \le conf \le 1.0\)	PASS
TC-16	ML Model	Monotonic timer profiling check	Latency > 0.0 ms	PASS
TC-17	Grip Controller	Initial instantiation	Status IDLE, force 0.0 N	PASS
TC-18	Grip Controller	Step force past 5.0 N threshold	Status CONTACT_DETECTED	PASS
TC-19	Grip Controller	Step force toward 50.0 N target	Enters STABLE_HOLD	PASS
TC-20	Grip Controller	Exceed 75.0 N safety limit	Clamped at 75 N; ABORT	PASS
TC-21	Grip Controller	Assert slip_detected = True	+18% boost; RECOVERY	PASS
TC-22	Grip Controller	Force above upper deadband (55 N)	Status DECREASING	PASS
TC-23	Grip Controller	Execute reset() command	Force=0.0 N, Status IDLE	PASS
TC-24	API Service	GET /health	HTTP 200; healthy	PASS

Test ID	Target Function	Test Condition & Input Vector	VerifiedAssertion	Status
TC-25	API Service	POST /predict valid payload	HTTP 200; prediction	**PASS**
TC-26	API Service	POST /predict malformed payload	HTTP 422 Unprocessable	**PASS**
TC-27	API Service	POST /calibration noise=0.08	HTTP 200; noise updated	**PASS**

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6.3 Machine Learning Metrics on Synthetic Dataset

The Random Forest classifier was evaluated on 144 holdout test samples and via 5-fold stratified cross-validation across 720 total synthetic samples:

- **Holdout Accuracy:** 100.0% (144 / 144 correct)
- **Macro-F1 Score:** 1.0000 across all 4 grip classes
- **5-Fold Cross-Validation Accuracy:** 100.00% ± 0.00%
- *Academic Disclaimer:* These metrics reflect synthetic signal separability, not clinical human EMG.

6.4 Host Telemetry & Latency Profiling

- **Minimum Measured Latency:** 1.40 ms
- **Mean Measured Latency:** 1.46 ms ± 0.08 ms
- **Maximum Measured Latency:** 1.88 ms
- **Projected ESP32-S3 Latency:** 3.1–5.3 ms (based on clock cycle analysis @ 240 MHz).

7. Closed-Loop Tactile Control Verification

The closed-loop force regulation dynamics were verified across five operating states:

1. **Free Travel Closing:** Actuator displacement advances by $(+3.5\text{ mm/cycle})$ while contact force remains nominal (< 5.0 N).
2. **Contact Threshold Detection:** At $F_n = 5.0\text{ N}$, the controller detects object touch and transitions from `CLOSING` to `CONTACT_DETECTED`.
3. **Proportional Tracking & Deadband:** Force steps upward at $(+1.5\text{ N/cycle})$ until entering the $[47.0, 53.0]\text{ N}$ deadband window, transitioning to `STABLE_HOLD` at 49.8 N.
4. **Autonomous Anti-Slip Reflex:** Upon asserting `slip_detected = True`, the controller executes an immediate +18% force boost (+12.0 N), arresting payload slippage in one cycle (20 ms).
5. **Anti-Crush Safety Ceiling:** Forced over-pressure is clamped strictly to 75.0 N, triggering `CRUSH_ABORT`.

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8. Second-Half Work Plan & Hardware Roadmap

With the software simulation prototype fully implemented, verified, and documented, the team's remaining work focuses on the five-phase hardware deployment roadmap:

Phase	Hardware Roadmap Objective	Key Engineering Tasks	Timeline
Phase 1	TinyML C Firmware Translation	Export RF trees to static C header tables via `emlearn`; flash to ESP32-S3 (240 MHz); benchmark embedded latency (< 5 ms).	Weeks 15–18
Phase 2	Analog Front-End (AFE) PCB	Design custom PCB with TI INA333 instrumentation amplifiers, 20–450 Hz bandpass filter, 50 Hz notch, and dry Ag/AgCl electrodes.	Weeks 19–22
Phase 3	3D Mechanical Hand Fabrication	3D-print anthropomorphic 5-digit chassis using carbon-fiber PETG/TPU; integrate coreless metal-gear micro-servos and Dyneema tendons.	Weeks 23–26
Phase 4	Tactile Array & Reflex Closure	Mold silicone fingertip pads embedded with Interlink 402 FSR sensors and micro-vibration slip accelerometers; close physical 50 Hz loop.	Weeks 27–30
Phase 5	Amputee Clinical Usability Trials	Obtain IRB ethical approval; recruit transradial amputees; conduct standardized functional tests (SHAP, Box & Blocks).	Weeks 31–36

9. Midterm Rubrics Evaluation & Approvals

Assessment Criteria	Max Marks	Awarded Marks
Problem Definition & Literature Review Depth	20	_____
System Architecture & Algorithmic Formulation	25	_____
Prototype Implementation & Code Quality	25	_____

Verification Evidence (Automated Tests, Build)	15	_____
Report Documentation & Presentation	15	_____
TOTAL SCORE	**100**	**_____**

_____ _____

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